

Background:

Recent research has shown that spontaneous behaviors play a critical role in shaping brainwide neural activity. Stringer et al. (2019), using large-scale two-photon calcium imaging in mice, demonstrated that ongoing neural activity is not mere noise but is structured and behaviorally driven. These findings challenge the traditional view of primary sensory cortices as purely stimulus-driven, suggesting instead that they integrate external sensory inputs with internal behavioral states.

Research question:

How do multidimensional spontaneous behaviors influence neural activity across different brain regions, and how does this interaction affect sensory processing?

Initial objectives:

- 1) To quantify the contribution of spontaneous behaviors to neural activity in various brain regions.
- 2) To determine the interaction between behaviorally driven neural activity and sensory stimuli.
- 3) To identify the specific behavioral variables that most significantly influence neural activity.

Methodology:

Initially, the Stringer dataset was retrieved, and various fields within it were analyzed by graphical analysis via graphs in python. Of these the parameters pupil area and running speed was chosen as the target variables. After this data pre-processing was done, by first z scoring the neural readings for each data field, and running PCA for 200 dimensions. UMAP was also run as an exploratory approach.

After this, first Neural networks of varying number of hidden layers and number of neurons per layer were applied, all of them with the Adam optimiser and with Mean squared error loss as loss function. Finally, GLM was used for the prediction of pupil area from neuronal data. L2 ridge penalty GLM was also tried.

Results:

The results showed that the Neural network models didn't work well, which could be due to a highly non linear relationship between the input vectors and the intended predictions, signifying the need to use alternate modes of learning and different architectures, including regularization. The GLMs on the other hand gave better results, but the GLM for running speed prediction unexpectedly degraded upon PCA pre-processing, which might be due to the nature of PCA as unsupervised, which diluted out the possibly weak variance between the I/O vectors.